

Denshin Impulse: Enhancing Connection with an Embodied Virtual Agent via Haptic Heartbeat Synchrony

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Figure 1: Overview of the Denshin Impulse experience. (A) The interaction scene: A participant, wearing a VR headset and a chest-mounted haptic device, makes arm contact with an adjacent physical doll, which represents the virtual agent’s physical body. (B) The VR experience: The participant’s first-person perspective within VR. (C) The core concept of Denshin Impulse: A symbolic representation of synchronized ECG-like waveforms originating from both the participant and the doll.

Abstract

We present “**Denshin Impulse**,” an interface that explores enhancing the connection with a virtual agent and promoting empathetic dialogue. This system presents synchronized heartbeats between a user in a virtual reality (VR) environment and the agent, mediated via haptic feedback from an adjacent physical doll. Our approach frames bilateral conveyance of shared cardiac rhythms as a user-facing expression of physiological synchrony often absent in human-agent interactions. As the user approaches the agent in the intimate zone, devices attached to the user’s chest and the corresponding physical doll deliver increasingly synchronized and intense haptic pulses. We expect to deepen the human-agent connection via this shared physiological sensation.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Virtual reality*.

Keywords

Virtual Reality, Embodied Conversational Agent, Physiological Synchrony & Heartbeat

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1 Introduction

We present “Denshin Impulse,” an interface that explores enhancing the connection with a virtual agent by conveying a shared heartbeat, embodying the Japanese concept of “Denshin (伝心)” for heart-to-heart communication. Large Language Models (LLMs) can simulate emotional understanding in empathetic conversations, often perceived as superior to human interactions. This capability suggests potential applications in fields such as education and psychotherapy, where empathetic communication is essential [Schlegel et al. 2025]. LLMs can be embodied as virtual agents and physical robots, and conveying emotions requires not only language but also non-verbal cues, such as facial expressions and body movements. The heartbeat, strongly linked to emotion (e.g., an accelerated pulse during anxiety[Hecquard et al. 2023]), serves as a non-verbal cue for

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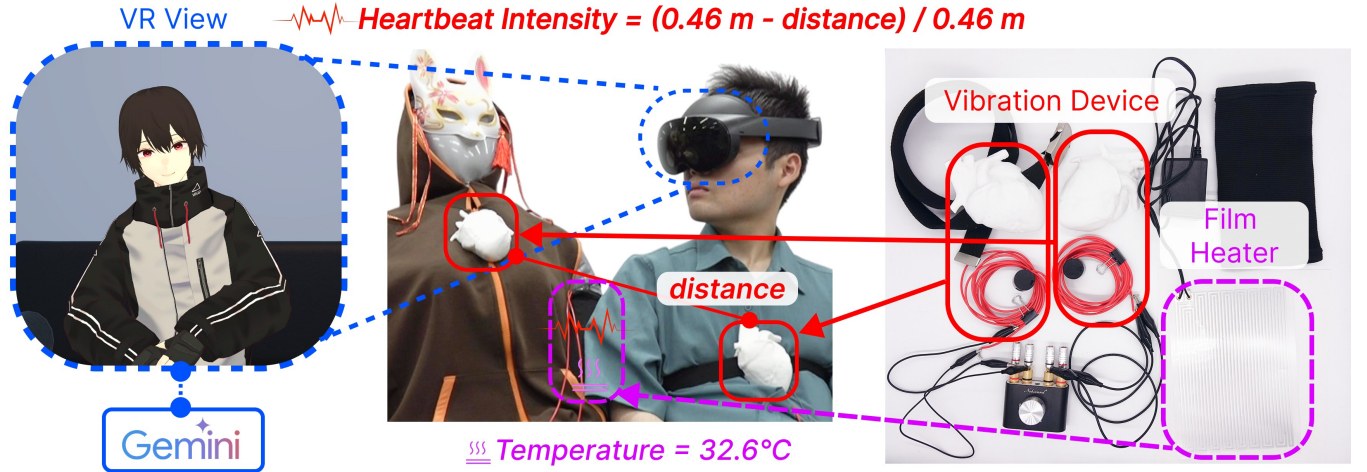


Figure 2: System architecture of “Denshin Impulse.” (Left) The virtual agent is rendered within VR; dialogue capabilities are facilitated by connection to an LLM (e.g., Gemini). (Center) The distance between the user’s avatar and the virtual agent in VR modulates the heartbeat intensity. The doll’s arm is maintained at a skin-approximating temperature (32.6 °C) by a film heater. (Right) The vibration actuator for haptic heartbeat presentation and the film heater for thermal feedback.

conveying emotions. Indeed, vibrational stimuli representing heartbeats can convey intimacy in interpersonal communication [Janssen et al. 2010]. However, a virtual agent’s heartbeat using a vibration actuator failed to elicit empathy when conveyed as a stress-induced elevated heartbeat [Hecquard et al. 2023] and presenting only the agent’s heartbeat was perceived as an intrusion into personal space, thereby increasing user discomfort [Ooms et al. 2025]. We posit that unidirectionality stems from the absence of “physiological synchrony. This phenomenon is the tendency in intimate human communication for individuals’ states to synchronize, including not only facial expressions but also physiological states like heart rate and breathing [Palumbo et al. 2017]. For instance, a strong sense of connection can foster self-disclosure in psychotherapy or provide companionship for lonely individuals.

We propose Denshin Impulse as an expression that manifests physiological synchrony between a user and an agent. Our contribution emphasizes a proxemics-driven expression, shared heartbeats that intensify as the participant and agent enter the intimate zone. In this demonstration, participants engage in dialogue to foster a connection with an agent within VR while maintaining physical contact (touching arms) with a life-sized physical doll seated beside them. As the distance between the participant and the agent in VR decreases, the synchronized heartbeats presented by the device attached to the participant’s chest and the doll’s chest become stronger.

2 System Design

Figure 2 illustrates the system architecture. We first describe the design for heartbeat control, followed by the design of the physical device. Heartbeat intensity is controlled by the distance between the user and the agent in the VR scene. According to Hall’s proxemics theory [Andersen et al. 2010], distances within 0.46 m are referred to as the “intimate zone,” a distance at which physical contact, permissible only in very close relationships, occurs. We control the heartbeat intensity to be minimal (0) when the distance between

the user’s chest and the agent’s chest is 0.46 m, and maximal (1) at 0 m. To generate the heartbeat vibration pattern, we created an animation that mimics an electrocardiogram (ECG) graph [Ooms et al. 2025]. Our physical device comprises: ACOUSTICAPTIC 639897 (FOSTER, Japan) tactile transducers that present the respective heartbeats; an inflatable mannequin; and an aluminum foil film heater (Heat Lab. Corporation, Japan). Tactile transducers convert the audio waveform of a heartbeat sound into nuanced (non-monotonic) vibrations. The film heater can generate a temperature of 32.6 °C when supplied with at least 5V and 2.4A. It reaches over 30° C within 45 seconds and maintains a stable target temperature. This film heater is secured to the doll’s arm using an armband.

3 Demonstration at SIGGRAPH Asia

Participants can engage in real-time voice dialogue with an adjacent agent in the VR while experiencing synchronized heartbeats, which dynamically change with proximity and the agent’s emotional state during the conversation. Upon approaching the agent, participants can make physical contact with the doll’s arm, experiencing the warmth of its touch and perceiving the intensifying synchronized heartbeats. In our preliminary user tests, participants reported that the proxemics-driven intensification of the shared heartbeat was particularly effective in making the agent feel “alive.” Participants can select their preferred agent’s appearance and choose a voice from English, Chinese, or Japanese.

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